Nermine Moumen

Industrial Chemical Engineering Student

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EDUCATION

National Institute of Applied Sciences and Technology (INSAT) *2022–Present*

Final-year Industrial Chemical Engineering student, specialized in Process and Environment, with a broad foundation spanning Thermodynamics, Fluid Mechanics, Energy & Mass Balances, Renewable energy and Materials Science. Seeking a final-year internship focused on process simulation and techno-economic evaluation of sustainable energy conversion routes.

PROFESSIONAL EXPERIENCE

SEGULA Technologies – Electrochemical Battery Modeling Internship *06/2026–08/2026*

- Modeled a **Li-ion cell** (NMC811/Graphite-SiOx) using the Single Particle Model (SPM) implemented in MATLAB.
- Solved the governing PDEs using the **Finite Volume Method (FVM)** with Crank–Nicolson time integration.
- Optimized the mesh and diffusion coefficient values, reducing the RMSE error from **202.13 mV to 40.5 mV**.
- Extracted and processed **experimental discharge data** from a database to validate MATLAB and COMSOL results.

Control Union – Quality Control Internship *08/2025–09/2025*

- Analyzed lubricants for viscosity and heavy metals (**ICP-OES**).
- Prepared and interpreted **analytical reports** to verify compliance with ASTM D5185 and **ISO 17025** standards.

ETAP – Water Treatment and Petrochemistry Internship *07/2024–08/2024*

- Studied water treatment methods (**coagulation-flocculation, ion exchange**) for contaminant and corrosion control.
- Performed organic geochemistry techniques (GC, GC-MS, Rock-Eval pyrolysis) for hydrocarbon characterization.

ADDITIONAL EXPERIENCE

Cascade – Online Chemistry Teacher (Part-time) *2024–Present*

- Provide individualized online chemistry tutoring for French Terminale students, covering exam-level concepts.

ACADEMIC PROJECTS

Techno-Environmental Analysis of Ethylene Production: Steam Cracking vs Bioethanol Dehydration

- Simulated both routes in **Aspen Plus V14**, comparing reactor design, separation train, and **energy integration**, achieving a purified ethylene stream at **99.95 mol% purity**.
- Conducted a cradle-to-gate **Life Cycle Assessment** in OpenLCA using Aspen Plus outputs as inventory data.

Prediction of Liquid–Liquid Equilibria and Cross-flow and Countercurrent Extraction for Ternary Systems

- Formulated the LLE thermodynamic system and solved it via Rachford-Rice and **arc-length continuation**.
- Reproduced 7 Mathematica notebooks proving equivalence of the Hunter-Nash and McCabe-Thiele methods.
- Reimplemented workflow in Python, MATLAB, and Mathematica (spline/fsolve, SciPy/mpltern, Interpolation/FindRoot).

Ethylene Oxide Production Process Simulation using Aspen HYSYS

- Simulated the ethylene oxide production process using **NRTL-RK** thermodynamics and **Langmuir-Hinshelwood kinetics**.
- Optimized operating conditions via **sensitivity analysis**, achieving 96% recovery and **97.14% final purity**.

PEM Fuel Cell and Electrolyzer Modeling with COMSOL and ANSYS Fluent

- Simulated isothermal and nonisothermal **PEMFC** behavior in COMSOL Multiphysics, achieving a cell voltage of 0.584 V.
- Modeled a 3D **PEM electrolyzer** in Ansys Fluent, achieving **1.40 L/h H_2 production** and identifying the oxygen evolution reaction as the dominant kinetic bottleneck.

TECHNICAL SKILLS

- **Simulation:** Aspen Plus, Aspen HYSYS, COMSOL Multiphysics • **Programming:** *MATLAB, Mathematica, Python*
- **Data & Office:** Microsoft Office Suite, OpenLCA • **Standards:** ISO 9001, ISO 14001, ISO 45001, ISO 17025, ISO 14040
- **Lab Techniques:** Chromatography (HPLC, GC, TLC), Spectrometry (MS, ICP-OES), Spectroscopy (NMR, FTIR, UV-Vis), Diffraction (XRD), Microscopy (SEM, TEM), Electrochemistry (Cyclic voltammetry, Polarography).

EXTRACURRICULAR ACTIVITIES

- **IEEE INSAT** – Chair of EMBS; Organizing Chair of TRSYP 3.0; Social Activities Chair of MENASYP.
- **THEATRO INSAT** – Member of the dance team.

LANGUAGES AND CERTIFICATES

Languages

English (Fluent), French (Fluent), Arabic (Native).

Certificates

Aspen HYSYS – MATLAB